

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	01-10-2026	Duration:	60 minutes

Code of Conduct

I Koppala Navya– 192511231 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Koppala Navya

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. **Propose a Suitable Solution Using Software Engineering Principles**

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Recommend safe and efficient routes based on real-time and historical conditions.

1. Understand the Problem Statement

1.1 Explain the problem in your own words.

People may face difficulties accessing **roads, public transport, hospitals, schools, markets, government offices, and other essential services**, especially in rural, remote, or poorly connected areas. Lack of proper roads, limited transport facilities, long travel distances, and unequal distribution of services can affect people's daily lives.

The proposed system aims to **assess accessibility using location-based and infrastructure-related data** and identify areas where access to roads, transport, and essential services is inadequate.

1.2 Identify the Objectives of the proposed system

- Assess the accessibility of roads and transportation facilities.
- Identify areas with poor connectivity.
- Measure the distance and travel accessibility to essential services.
- Identify underserved or difficult-to-reach locations.
- Provide useful maps, indicators, and reports for decision-making.
- Support better planning of roads, transport facilities, and essential services.

1.3 State the Scope of the problem

- Road network accessibility analysis.
- Public transport accessibility assessment.
- Accessibility to hospitals, schools, markets, and government services.
- Geographic/location-based analysis.
- Identification of underserved areas.
- Visualization using maps, charts, and accessibility indicators.

- Can be extended to real-time transport and traffic data.

2. Identify Key Stakeholders and Challenges

2.1 Identify All the Stakeholders Involved in the System

The main stakeholders involved in “Assess the Accessibility of Roads, Transport Facilities, and Essential Services” are:

1. Citizens / Residents
2. Local Government
3. Transport Authorities
4. Urban/Rural Planning Departments
5. Emergency Services
6. Essential Service Providers
7. System Administrators

2.2 Explain the Roles and Responsibilities of Each Stakeholder

Stakeholder	Roles and Responsibilities
Citizens / Residents	Use roads, public transport, hospitals, schools, markets, and other essential services. Provide feedback about accessibility problems.
Local Government	Plan and improve roads, transport facilities, and public services using accessibility information.
Transport Authorities	Manage public transport routes, stops, schedules, and transportation availability.
Urban/Rural Planning Departments	Analyze infrastructure gaps and identify areas that require better connectivity and facilities.
Essential Service Providers	Provide healthcare, education, markets, and government services and maintain service-related information.
System Administrators	Manage system data, users, updates, security, and overall system operations.

2.3 Major Challenges and issues faced by the stakeholders

- Poor road connectivity in some areas.

- Limited public transport availability.
- Long distance to essential services.
- Unequal distribution of facilities.
- Traffic congestion and travel delays.
- Incomplete or outdated geographical data.
- Difficulty identifying underserved locations.
- Lack of a centralized accessibility analysis system.

3. Analyze the Current Approach

3.1 Describe the existing system or current process.

Currently, accessibility is often assessed using **individual maps, surveys, government records, spreadsheets, and manual analysis**. Different departments may maintain separate information about roads, transport facilities, hospitals, schools, and other services.

3.2 Identify its strengths and limitations.

Existing maps provide basic geographical information.

- Government departments already maintain infrastructure data.
- Field surveys can provide real-world information.
- Existing mapping tools can visualize locations.

3.3 Analyze the problems or gaps that need to be addressed.

The major gap is the absence of an integrated software system that combines road, transport, and essential-service information to calculate and visualize accessibility for different areas.

For “**Assess the Accessibility of Roads, Transport Facilities, and Essential Services**”, the following problems and gaps need to be addressed:

1. **Poor Road Connectivity** – Some rural and remote areas have inadequate or poorly connected roads.
2. **Limited Public Transport** – Certain locations may have fewer buses, transport routes, or transport facilities.
3. **Long Distance to Essential Services** – People may need to travel long distances to reach hospitals, schools, markets, and government offices.
4. **Unequal Distribution of Services** – Essential facilities may be concentrated in some areas while other locations remain underserved.
5. **Traffic Congestion and Travel Delays** – Congestion can increase travel time and reduce accessibility.

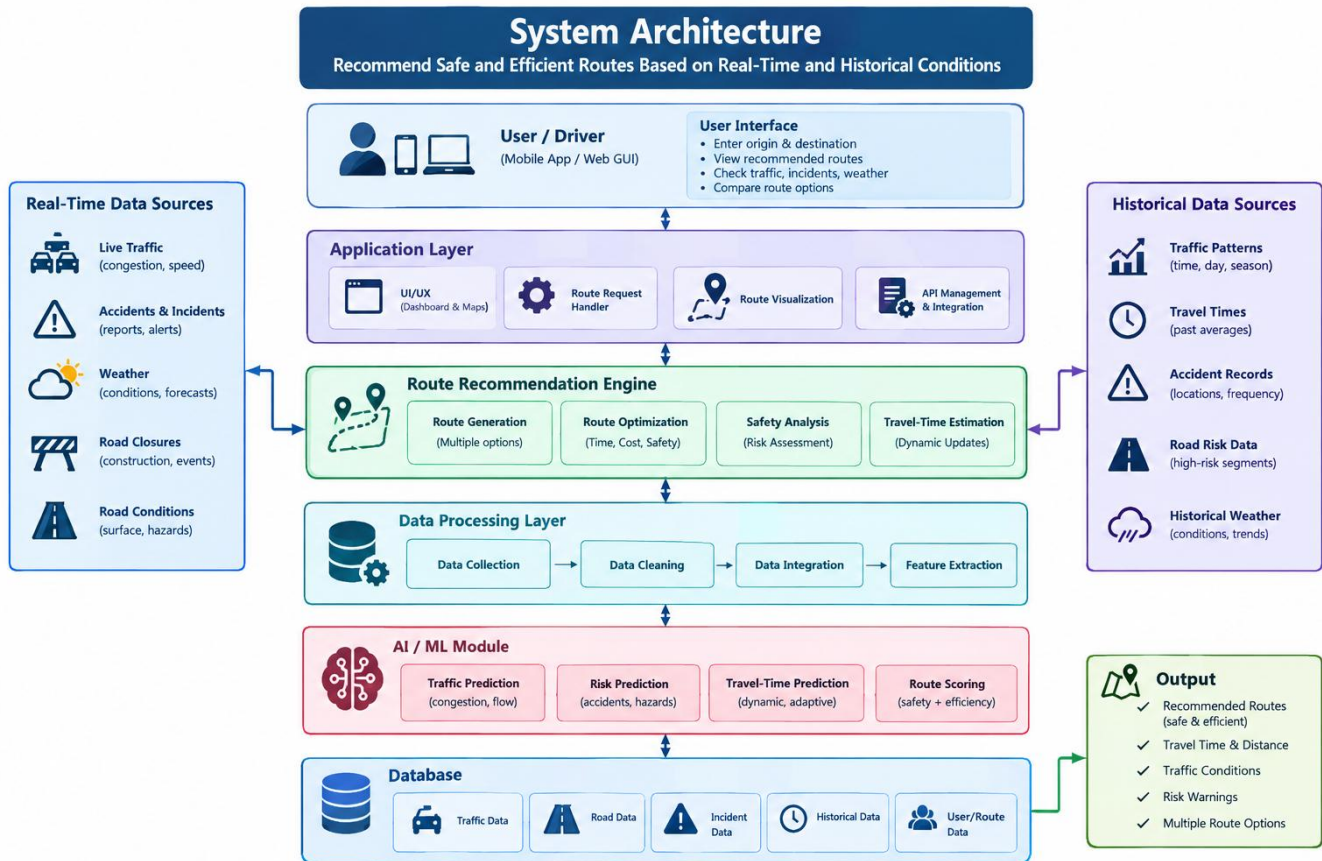
4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

A **Location-Based Accessibility Assessment System** can be developed to collect and analyze geographical information about roads, transport facilities, and essential services.

The system can calculate accessibility using factors such as:

Distance + Travel Time + Road Connectivity + Transport Availability + Service Availability



4.2 Describe the major functional modules of the proposed system.

1. Data Collection Module

- Collect road network data.
- Collect public transport locations and routes.
- Collect hospitals, schools, markets, and other service locations.
- Store geographical coordinates and related information.

2. Road Accessibility Module

- Analyze road connectivity.

- Calculate distance between locations.
- Identify poorly connected areas.
- Analyze road availability and coverage.

3. Transport Accessibility Module

- Identify bus stops, railway stations, and other transport facilities.
- Analyze distance to transport facilities.
- Assess availability of transportation options.
- Identify areas with limited transport access.

4. Essential Services Accessibility Module

- Analyze accessibility to hospitals.
- Analyze accessibility to schools.
- Analyze accessibility to markets and government services.
- Identify locations that are far from essential facilities.

5. Map and Report Module

- Display results on interactive maps.
- Use markers and accessibility zones.
- Generate charts and reports.
- Support infrastructure planning and decision-making.

4.3 Identify the functional and non-functional requirements.

User should be able to enter or upload geographical data.

- System should store road and service locations.
- System should calculate distances and accessibility.
- System should identify underserved areas.
- System should display results on maps.
- System should generate reports and charts.
- Administrator should be able to update system data.

Non-Functional Requirements

- **Usability:** Interface should be simple and easy to understand.
- **Performance:** Accessibility calculations should be completed efficiently.
- **Scalability:** System should support increasing geographical and service data.
- **Security:** User and system data should be protected.
- **Reliability:** Results should be generated consistently.
- **Maintainability:** Modules should be easy to update.
- **Availability:** System should be accessible whenever required.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

Modularity → System is divided into independent modules.

Scalability → New locations, services, and geographical datasets can be added.

Maintainability → Individual modules can be modified without affecting the entire system.

Reliability → Validated data and consistent algorithms improve the reliability of results.

Usability → Maps and visual reports make the results easy to understand.

Security → Authentication and controlled access can protect system data.

5. Justify the Chosen Methodology / Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

Selected SDLC Model: Agile

The **Agile software development methodology** is suitable for this system because the requirements and geographical datasets may change during development.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

- Development can be divided into small iterations.
- Individual modules can be developed and tested separately.
- Feedback can be collected from users and stakeholders.
- Changes in accessibility requirements can be incorporated easily.
- Maps and visualization features can be improved continuously.
- Problems can be identified early through regular testing.

5.3 Explain how it supports successful project development and maintenance.

Requirement Analysis

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Module Planning

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Data Collection & Design

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Incremental Development

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Testing



User Feedback



Improvement



Deployment & Maintenance

Support for Maintenance

Agile supports continuous maintenance because new roads, transport facilities, and essential services can be added to the system through future development iterations.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

The proposed system is expected to:

- Identify areas with poor road accessibility.
- Identify locations with limited public transport.
- Measure accessibility to hospitals, schools, markets, and other essential services.
- Highlight underserved areas through maps.
- Reduce the effort required for manual accessibility analysis.
- Provide centralized accessibility information.
- Support infrastructure and transportation planning.
- Help authorities prioritize areas requiring improved connectivity.
- Provide visual reports for easier decision-making.

6.2 Explain how the solution addresses the identified challenges.

Challenge	Proposed Solution
Poor road connectivity	Road network accessibility analysis
Limited transport	Transport facility and route analysis
Long distance to services	Distance-based service accessibility analysis
Unequal service distribution	Identify underserved areas
Manual analysis	Automated accessibility calculations
Distributed information	Centralized accessibility platform
Difficult visualization	Interactive maps and charts
Changing infrastructure	Regular data updates

6.3 Discuss potential risks, implementation challenges, and limitations.

- Inaccurate or outdated geographical data.
- Difficulty obtaining complete road and transport datasets.
- Internet/GPS dependency for location-based features.
- Large datasets may require higher processing capacity.
- Accessibility calculations may not capture every real-world condition.
- Changes in traffic and transport schedules can affect actual travel time.
- Maintaining updated infrastructure information can require continuous effort.

Limitations

- Results depend on the accuracy and completeness of input data.
- Basic distance calculations may not represent actual travel conditions.
- Temporary road closures or transport disruptions may not be reflected immediately.
- The system may initially cover only a selected geographical region.
- Social and economic factors affecting accessibility may not be fully represented.

6.4 Suggest possible future enhancements.

- Add real-time public transport tracking.
- Include weather and road-condition information.
- Use **AI/ML** to predict accessibility problems.
- Develop a mobile application for citizens.
- Add emergency-route accessibility analysis.
- Include wheelchair and differently-abled accessibility.
- Integrate live GPS and IoT-based infrastructure data.
- Provide automatic alerts for newly identified underserved areas.
- Expand the system to larger cities, districts, or regions.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.

Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing;
	strong technical justification.			report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5

TOTAL	<i>/25</i>
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